

LSI-lite: A Composition Analysis Tool

Profiled Composition Structural Exploration (Δx , r_v , ρ_r)

What It Does

LSI-lite measures how an image **behaves under compositional structure** using three primitives: Δx (off-center gravity), r_v (void ratio), ρ_r (rupture/mark energy)—and tells you if it sits within intended bands for its class. It's built to study **stability and failure** as you iterate, not to crown winners.

- **Balance** (Δx): How far the visual center is from the geometric center
- **Density** (r_v): The ratio of empty space to filled space
- **Detail** (ρ_r): The amount of edge energy and texture density in key areas

It combines these measurements into a 0-100 score and tells you whether an image "passes" basic compositional criteria structural exploration test.

Why researchers use it (not FID/CLIP/SSIM).

Resemblance and caption metrics can't answer: *will this composition hold* when pushed? LSI-lite is a **composition structural exploration test**—a quick, profiled gate you can log across Baseline → Pressure → Collapse-trigger runs.

The Three Measurements

Δx (Delta-x): Off-Center Balance

- Finds the visual center of mass using edge detection
- Measures how far it is from the geometric center
- Range: 0.0 (perfectly centered) to 1.0 (at edge)
- **Why it matters:** Dead-center compositions often feel static

r_v (Void Ratio): Empty Space

- Calculates the percentage of background vs foreground
- Higher values = more empty space
- Range: 0.0 (completely filled) to 1.0 (mostly empty)
- **Why it matters:** Negative space creates visual breathing room

ρ_r (Rho-r): Rupture Energy

- Measures texture and detail density around the main subject
- Uses Laplacian edge detection over a "halo" around the subject
- Range: 0.0 (smooth) to 1.0 (highly detailed)
- **Why it matters:** Controls visual complexity and focus

Three Profiles

Different types of images need different compositional rules, this lite version covers:

Figure_Default

- **For:** Portraits, figure drawings, character art
- **Emphasis:** Balance (45%), then density (35%), then detail (20%)
- **Expects:** Moderate off-centering, balanced space usage, controlled detail

MarkMaking_Expressive

- **For:** Sketches, expressive artwork, gestural drawings
- **Emphasis:** More balanced weights (40%/30%/30%)
- **Expects:** Looser centering rules, higher texture tolerance

Landscape

- **For:** Landscape photos and paintings
- **Emphasis:** Density matters most (40%), then balance (35%), then detail (25%)
- **Special feature:** Detects water reflections and adjusts balance measurement accordingly

Gate rule (accept/reject).

Scale **K_lite** (weighted geometric mean of in-band primitive scores) to **LSI_lite_100**. Accept if **LSI_lite_100** $\geq$ 55 and no **RED bands**. Otherwise reject—and log which primitive is limiting (the **priority_knob**).

How Scoring Works

1. **Measure:** Calculate all three primitives (Δx , r_v , ρ_r)
2. **Check bands:** Each primitive has an acceptable range - values outside are flagged "RED"

3. **Score:** Combine the three values using weighted geometric mean
4. **Gate:** Image "passes" if score ≥ 55 AND no RED flags

Example Output

What a single run gives you (at a glance).

- **Read card:** profile • $\Delta x/r_v/p_r$ (0–1) • per-band **OK/RED** • **LSI_lite_100** • **accepted** True/False • **priority_knob** (+ suggested increase/decrease).
- **Stamp:** **ACCEPT / REJECT / UNKNOWN** with a one-line reason.
- **Exports:** a composite PNG + one CSV/JSON row (fields below).

What it is *not*.

Not aesthetic judgment. "Accepted" means "within intended structural bands," not "good art." Profiles reflect specific design intent; many excellent images will fail on purpose.

What This Tool Is Good For

- **Art education:** Consistent feedback on basic composition principles
- **Batch analysis:** Evaluate large sets of images quickly
- **AI research:** Study how different generation methods handle composition
- **Quality filtering:** Screen images for basic structural soundness

Important Limitations

Not Universal

- Built around Western compositional conventions
- Many excellent artworks will fail these tests intentionally
- Abstract art, experimental compositions may be penalized unfairly

Not Aesthetic Judgment

- "Accepted" means "follows basic rules," not "good art"
- Measures structure, not beauty, emotion, or meaning
- Can't evaluate color, concept, or cultural significance

Current Scope

- Three primitives can't capture all compositional elements
- Profile bands are designer intuition, not empirically validated
- Works best on clear subject/background images

Technical Notes

- Handles EXIF rotation automatically
- Resizes to 1536px longest side for consistency
- Uses bilateral filtering and morphological operations for robustness
- Fallback handling for edge cases (no foreground detected, etc.)

Research Context

LSI-lite is designed as a research instrument to study compositional stability in AI-generated images. Unlike metrics that measure similarity to training data (FID) or text-image alignment (CLIP), this focuses on structural principles that observe how compositions hold up under iteration or modification.

The goal is not to replace human artistic judgment, but to provide a systematic way to discuss why some compositions feel more stable than others.

Artist's Statement

Structural Exploration. Artists don't march images through single images or a linear test; we explore a space of alternatives. Pressure is applied as a contradiction (e.g., torque↑ & permeability↑). The image may collapse, re-cohere, or discover a **new center**. LSI-lite doesn't score "beauty"; it logs whether off-center gravity (Δx), breathing ground (r_v), and mark energy (p_r) support the path you're taking. The guardrails aren't grades, they're **questions**: where does gravity sit, do voids breathe, are marks committed? Exploration ends when form is found or the gap is named.

How LSI fits that story

- **Δx (gravity)** → “Did we escape centering in a way that reads as intentional gravity, not tip?”
- **r_v (voids)** → “Are the alternatives ventilating, or sealing?”
- **p_r (rupture)** → “Is energy committed, or decorative noise?”
- **Gate (≥ 55 , no RED)** → “Is this branch structurally sound *for this profile*?”
- **Priority knob** → “If we pursue this branch, what single nudge makes it truer to itself?”

Non-linear path (what we actually see)

- **Default → Collapse → Cohesion → New Center**
or
- **Default → Deviation → Poise (hold) → Deconstruction (on purpose)**
LSI marks the **state changes**, not a pass/fail morality.

*It's fieldwork in a landscape of alternatives. Each image is a branch; pressure asks whether the branch **finds form** or collapses into defaults (human and model). LSI isn't judging taste; it's the start of a **compass** that tells you where structure holds while you deviate.*

This is an experimental tool. Use it to explore patterns in images, not as a definitive judgment of artistic quality.

Appendix:

Methods (how it works, reproducible)

Primitives (what/why/how)

- **Δx — Centroid offset** (*off-center gravity / frame tension*)
How: bilateral denoise → **Canny(50,100)** → **edge centroid**; **fallback** to foreground centroid if edges empty. If both fail ⇒ **$dx = \text{NaN}$** (mark **unknown**, gate-fail).
- **r_v — Void ratio** (*figure-ground breath / negative space*)
How: Otsu threshold → largest component → open/close → **$rv = 1 - \text{fill}$** .
- **p_r — Rupture energy** (*mark/texture pressure*)
How: **Laplacian $k=3$** → abs → normalize → **mean over subject + halo** (halo fraction per profile).

Profiles (guards & masks)

Each profile specifies weights and **band guards** (min..max). **Figure/MarkMaking** use a subject halo (0.08 / 0.12).

Landscape uses **$\rho_{\text{mask}}=\text{full}$** and may apply **$\Delta x$ ROI** via a reflection probe (**$\text{top_frac}=0.60$**) *for Δx only*. ** r_v/p_r are always full-frame.*

Scoring

- Per-primitive **Gaussian in-band score** (0–1) from guard range.
- **K_{lite}** = weighted geometric mean of the three soft scores with **$\epsilon=1e-6$** floor.
- **$LSI_{\text{lite}}_{100} = 100 \times K_{\text{lite}}$** .
- **Accept** iff **$LSI_{\text{lite}}_{100} \geq 55$** and all bands **OK** and not **unknown**.

Quality-of-life diagnostics (for iteration)

Alongside $\Delta x/r_v/p_r$:

$*_{\text{center}}$ (band midpoint) • **$*_{\text{to_center}}$** • **$*_{\text{to_center_norm}}$** (0..1) • **$*_{\text{direction}}$** (“increase/decrease”) • **priority_knob** (which primitive is farthest from center) • **priority_score** .

Unknown semantics

If edges and foreground both fail: **$dx = \text{NaN}$** , **$\text{unknown}=\text{True}$** , bands = RED, gate-fail (this prevents silent “passes” on blank/edgeless frames).

Run loop (single pass)

Load once (EXIF-aware) → compute $\Delta x/r_v/p_r$ **once** (thread the **Δx ROI** if Landscape probe triggers) → assemble read card.
Never recompute/overwrite Δx .

I/O and types

Masks are `uint8`; don't truth-test NumPy arrays (`if arr:`). Use `is None`, `np.any`, `np.all`. Runs local or Colab (no hard pins unless you need exact repro).

What you get: Visualizations & Artifacts

Single-image visualization (3×3 grid)

1. Original (blue = geometric center; red = Δx centroid)
2. Grayscale used for analysis
3. Foreground mask (fill %)
4. Edges (Canny 50/100)
5. Halo mask preview (for p_r)
6. Laplacian energy heatmap (p_r field)
7. **Band gauges** ($dx/r_v/p_r$): guard bar + center tick + measured marker + “increase/decrease”
8. **Gaussian curves** (tiny sparklines per primitive) with current value plotted
9. **Acceptance stamp** (**ACCEPT** / **REJECT** / **UNKNOWN**) + one-line reason
Optional overlays: reflection ROI (Δx only), value histogram.

Exports:

- **PNG** of the grid
- **CSV/JSON row** per image with this schema:

```
profile,dx,rv,rho,band_dx,band_rv,band_rho,LSI_lite_100,accepted,unknown,
dx_center,rv_center,rho_center,
dx_to_center,rv_to_center,rho_to_center,
dx_to_center_norm,rv_to_center_norm,rho_to_center_norm,
dx_direction,rv_direction,rho_direction,
priority_knob,priority_score,dx_roi_used
```

Batch visualization (dataset)

- **Score histogram** (+ gate line @55, acceptance rate)
- **Band failure breakdown** (% RED by $dx/r_v/p_r$)
- **Priority-knob distribution** (what limits most, and direction $\uparrow/\downarrow$)
- **Δx vs r_v scatter** (colored by LSI; target-zone overlay)
- **Correlation heatmap** (Δx , r_v , p_r , LSI)
- **Box/violin by profile** (primitive distributions across Figure / MarkMaking / Landscape)
- **Seed-stability view** (variance across seeds for the same prompt)
- **Failure atlas tiles** (top examples per Category-Ledger tag)
- **Systematic Gap table** (prompt → acceptance → common failure → gap note)

Use in studies (how to report, minimal)

- **Profile** used + **gate rule** (≥ 55 and no RED).
- **Acceptance rate**, **band-failure mix**, and **priority-knob distribution**.
- **Seed stability** for repeated prompts.
- **Systematic Gap** notes when language lacks a visual schema (optional but recommended).

One-liner (Methods box):

We use LSI-lite, a profiled composition structural exploration that scores off-center gravity (Δx), void ratio (r_v), and rupture energy (p_r) against profile band guards. A weighted geometric mean (ϵ -floored) yields `LSI_lite_100`; an image passes if `LSI_lite_100` ≥ 55 and all bands are OK; edgeless/foregroundless frames are flagged unknown and fail. Profiles: `Figure_Default`, `MarkMaking_Expressive`, `Landscape`; a reflection probe may apply a Δx ROI in Landscape only. LSI-lite_Compositional_Analysis

Limits

- **Diagnostic, not aesthetic.** “Accepted” $\neq$ “good art.”

- **Profiled, not universal.** Western composition bias; many great images will fail **on purpose**.
- **Scope.** Color, concept, cultural context are out-of-scope; three primitives can't capture all structure.

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Authorship

This framework was architected by Russell Parrish and recursively co-developed inside GPT, Gemini and Claude. Every critique is human-led; every recursion is model-driven. The result: a reasoning layer authored through language, not image manipulation.